# Native projective synthesis for scalar-local Eisenstein digits

**Question.** Does constructing selected 31+32τ digit points directly in native projective coordinates improve the scalar-local path, and what correctness evidence covers the measured and full-fixture workloads?

**Scope and instance.** This is an engineering study of SEC 2 secp256k1 (E: $y^2=x^3+7$, $p=2^{256}-2^{32}-977$, subgroup order n=FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFEBAAEDCE6AF48A03BBFD25E8CD0364141). The fixture contains 12,288 scalar/point pairs. Its SHA-256 is:

a1c4964f174e36ea26f6a1897128f651a2ce7251e4bf4ffd532ff09221e721d3

It does not add a curve, isogeny, scalar rule, or universal correctness theorem.

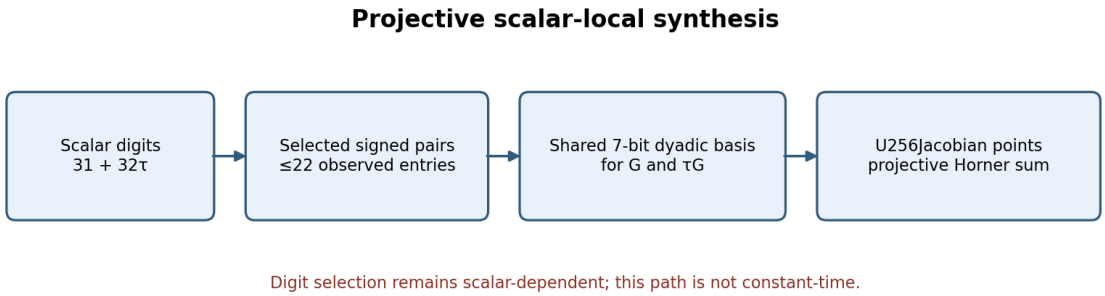



Figure 1: Projective digit synthesis pipeline

**Implementation review.** The hash-guarded native overlay replaces affine point synthesis from the selected signed digit pairs with a shared seven-bit dyadic basis for G and τG. Each selected pair is assembled using native U256Jacobian operations, and the Horner accumulator consumes those projective points directly via add_projective. This avoids the earlier pair-field window construction and affine conversion. The frozen manifests observed at most 22 selected entries; this is not a peak-memory measurement.

The selected pairs, map lookups, synthesis loop, and additions still depend on scalar digits. The implementation is therefore **not constant-time** and is not suitable for secret-scalar use. No side-channel assessment or peak-memory measurement was performed.

**Matched Tier 1 panels.** The one-scalar and 128-point v2 manifests compare projective synthesis against the preceding scalar-local builder on identical frozen inputs. Both ran on RunPod pod thyvmdqmzsz5ik, CPU 40, with 16 alternating repetitions, 96 completed arms, 48 completed pairs, and adjacent reference/reference A/A controls. The Rust Instant interval covers scalar-local preparation plus evaluation; process launch, fixture loading, and external replay are outside the interval. The two manifests and 128-point manifest, their rules, launches, and archived raw outputs identify job 508270ee2b714b2db002f9c37b906f38; corresponding 128-point launch and raw outputs identify job 47fb3c09560f494eb27769d25135e7e4. The common native provenance pins the sources and executable.

| Frozen workload | Earlier builder mean | Projective mean | Qualified paired median | Paired bootstrap 95% interval |
|---|---|---|---|---|
| One scalar, cold | 3.199 ms | 0.543 ms | 5.916× | [5.853, 5.937] |
| 128 points, cold | 349.468 ms | 39.829 ms | 8.761× | [8.719, 8.802] |

The independent panel auditor verified frozen source/binary and raw archive hashes, commands and outputs, the pair ledgers, A/A gates, and independent replay of all measured outputs. Both comparisons

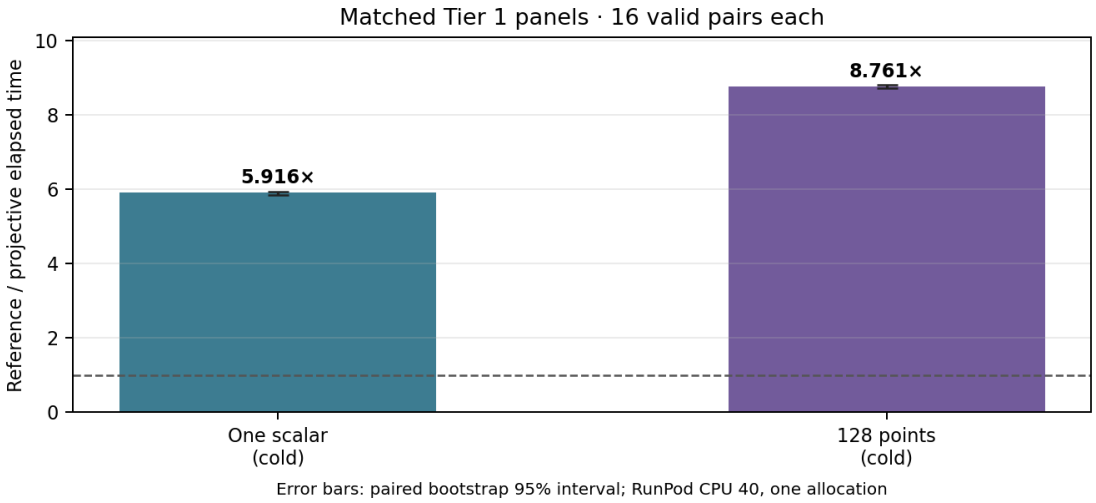



Figure 2: Matched projective timing ratios

were qualified, artifact-stable, and had no invalid arms or A/A failures. These are Tier 1 observations for one allocation and the specified cold workloads, not a general performance guarantee.

**Dispatch correction retained.** The first manifests accidentally dispatched the earlier scalar-local executable in both comparison slots. The preserved jobs `6815d8a260564a7ba4bbd31d166a969e` and `844d71a000a94f0dafc1dc9a6b9f54a1` are valid A/A controls only; the v1 audit receipts classify them as `invalid_plan_candidate_equals_reference` and leave the speedup null. The corrected v2 plans were frozen before dispatch and name the projective candidate. No data from the v1 controls is promoted as a performance result.

**Full-fixture correctness replay.** The separate serial manifest, launch, and raw archive preserve job `3cddff74cf9349bdaec7b6910c4884a0`. Both identical candidate arms verified all 12,288 outputs with stable artifacts; the auditor also checked the frozen fixture digest, binary, raw commands and output, pair ledger, and independent final-scalar replay. The arms took 3.850 s and 3.854 s. This is an A/A correctness and cost-shape replay with no speedup claim; it does not establish correctness for all scalars.

**Graph and proof status.** The study concerns an implementation change for the existing secp256k1 instance, so it creates no new canonical curve/isogeny graph edge or node. I checked the experiment's adaptive digit graph record, the earlier scalar-local report, and the repository's canonical candidate catalog and route records; none is changed by this result. The finite replay does not prove the all-scalar group-law refinement. The shared sparse-radix correctness problem tracks the unresolved selector and native-to-mathematical bridge, with its acceptance criteria and next discriminating checks.

The editable figure source, SVG and 180-dpi PNG copies, and this report's PDF are committed together. The PDF includes the report and both visuals.